

Fig. 4: Author's prototype

from its internal oscillator. Base current of TUS (T1) is varied from 0µA to 175µA in eight steps using port D pins (RD0, RD1 and RD2) of IC1. Collector voltage of T1 is varied from 0V to 2.5V in eight steps using IC1 and IC3 through IC4 and T2. Output voltage from the DAC is amplified about four times using LM358 and given to the base of T2.

Emitter of T2 is connected to the collector of TUS, whose voltage is being varied. Collector voltage of TUS is

given to channel-0 (AN0) of the built-in analogue-to-digital converter (ADC) of IC1 after dividing the voltage by half using a potential divider arrangement (R19 and R20) to meet the voltage limitation of the microcontroller.

Assuming that the collector current of TUS is equal to its emitter current, output is taken across the 1-ohm resistance, and the drop across 1-ohm resistance is amplified about 50 times and given to channel-1 (AN1) of the built-in ADC of IC1.

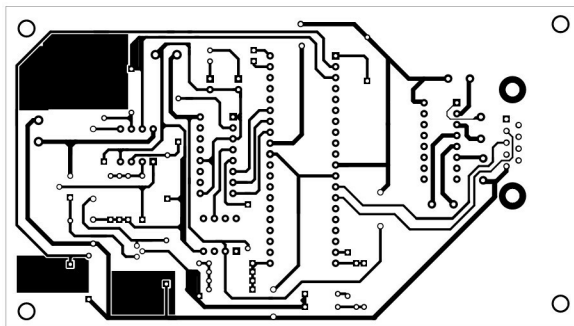


Fig. 5: Actual-size PCB layout of the serial interface using Python software

Construction

An actual-size, single-side PCB layout for the USB interface is shown in Fig. 5 and its component layout in Fig. 6. After assembling the circuit on the PCB, enclose it in a suitable box.

Firmware

The firmware is written in C and compiled using CCS C compiler. It implements the following tasks:

1. Generates different base currents
2. Generates different collector voltages
3. Measures the voltage and current (current in terms of voltage)
4. Sends the data to the PC

Host software

The application software to collect the measured value and display it on the monitor screen has been developed using Python. It requires the following components to be installed on your system: Python 2.5 (python-2.5.2) or higher IDE, Win32 Python (python32-210.win32-py2.5), Serial Port (pyserial-

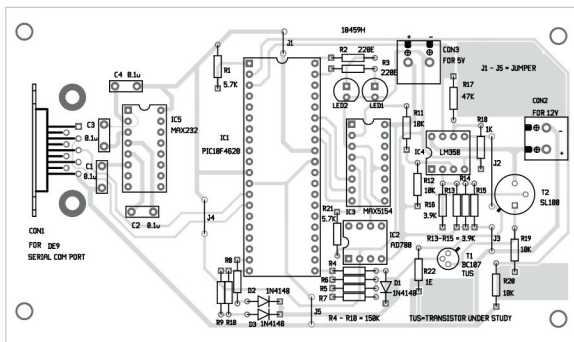


Fig. 6: Component layout of the PCB